Mathematical overview

Small witnesses: any point in the closed convex hull of a finite planar dot set lies in a hull of at most three original dots. Triangulating the convex boundary polygon proves this; degenerate point/segment hulls need at most one/two. Three is sometimes necessary. The printed T lies on no joining segment and has several containing triangles, so exactly three are needed. An interior target on a diagonal may need only two.

Separation: two disjoint closed convex hulls of finite nonempty groups admit a line placing them strictly on opposite sides. A perpendicular bisector of a closest-point segment works. Conversely a strict separator prevents the hulls meeting because open half-planes are convex. The printed AC/BD diagonal regions intersect and cannot be separated; AB/CD can. A thick line or a dot on the line does not meet the strict rule.

Three groups: their hulls must share one point, not merely meet pairwise. Five labels on a line always suffice: outer pair, next-inner pair and middle singleton share the middle location, even with coincidences. Four distinct line positions fail, so five is sharp. The regular hexagon succeeds with three opposite pairs; the irregular convex hexagon has no split. These examples do not establish the general seven-point planar Tverberg theorem.

Grades 2-3 can remove dots, place a ruler line and build groups. Grades 4-5 can explain triangle coverage, a separator obstruction and the exhaustive hexagon argument. Experiments find candidates; complete geometric certificates justify necessity and impossibility.

Entry routes and readiness

Shared Grades 2-5: retain that regions are filled and closed, with singletons as points and pairs as segments. Adult reading is fine. P1-2 discard dots intentionally; P4-6 use every label once in three nonempty groups. The three-group condition and an impossibility explanation are later gates.

Current fixed K–1 / 2–3 / 4–5 tables have one anchored adult each. Choose one investigation and its short launch per table per visit. Adults read or record while children own choices; pool cases into one checkable record. A K-1 oral target-retaining or ruler-separator entry is suitable only while the child retains the rule.

Materials and preparation

Per pair: twelve movable labeled dot markers, one target marker, three colors/patterns of pencil, six tracing sheets, a straightedge, blank paper, tape and erasers. Main boards are at least 130 mm square; print at 100% with equal scaling on both axes. Thick strokes and marker edges do not define the mathematical regions; use centers.

Show the filled hull, not only its outline. Tracing copies keep fixed-group attempts separate. Prepare labels/overlays in about 10-15 minutes. Physical registration, singleton interpretation and classroom rule retention remain untested.

Launch together

Let pairs move dots briefly. Use the printed dots-to-band-to-filled-region example, then remove one non-task dot and ask whether a target remains inside. Before three-group work, show what one point in all three means; three different pairwise meetings do not count. Keep the demonstration short and let children choose their own attempts.

First visit and return visits

P1-2 can take a 20-30 minute visit; P3 a shorter separator visit. P4 is the concrete three-group entry before P5-6's planar contrast. Those may occupy a further 20-35 minutes or separate return visits. Times are proposals. Pool line/subset checks and retain one target witness, separator or grouped drawing to resume.

Problem 1-2: Keep a target using the fewest dots

Entry, prerequisites and timing

Grades 2-3 can physically remove markers and redraw the filled region. Grades 4-5 can prove the at-most-three fact and audit no-pair evidence. Allow 20-30 minutes. The target remains fixed while supporting dots change.

Mathematical facts and solutions

P1 exactly three dots are necessary and sufficient. Adult coordinates are A=(1,0), B=(5,0), C=(8,2), D=(7,6), E=(3,7), F=(0,3), T=(16/5,17/5). None of the fifteen joining lines contains T, and it coincides with no dot, ruling out one/two. BEF contains T: T=(2/5)B+(2/5)E+(1/5)F, with positive weights summing to one. This is an exact adult containment check, not required child notation.

All containing triples are ABE,ACE,ADE,ADF,BDF,BEF,CDF,CEF. Any one is a valid answer. A child can show why two cannot work by pooling transparent pair-segment checks and keeping the complete collection checkable; every child need not redraw all fifteen.

P2 one-dot target: A itself. Exactly-two target: midpoint (3,0) of the exposed edge AB, different from every original dot. Exactly-three target: printed T. A merely interior point is not automatically a three-dot example; if it lies on a diagonal it can use two.

No target in the original filled hull needs four. Draw noncrossing diagonals from one boundary vertex to triangulate the convex hexagon. The triangles cover the whole closed polygon, and each uses three original dots. Any target lies in one of them. For an arbitrary finite dot set, use its boundary vertices; if the hull degenerates to a point or segment, one or two extreme dots suffice. This proves the planar at-most-three witness theorem.

Hold-back hints

First let pairs remove dots and check what changes. If two dots are proposed, draw the entire segment between them and locate T exactly. For the upper bound, ask how the large filled polygon could be covered by smaller regions using original dots.

Depth and further questions

Mark several different targets on one tracing copy. Boundary vertices need one dot, nonvertex edge points need two, and targets off every pair segment need three. A pair-segment arrangement partitions the possible targets into concrete cases without introducing barycentric arithmetic.

Facilitator notes

This is a small supporting subset, not a partition of every label. Keep the original hull on a separate copy so deleting markers does not move the target or erase the reference region.

Problem 3: A line that separates two fixed groups

Entry, prerequisites and timing

Grades 2-3 can try a ruler line and redraw each group's filled segment. Grades 4-5 can explain why the crossing defeats every line. Allow 15-25 minutes; reading is the only textual prerequisite.

Mathematical facts and solutions

The fixed square has A=(1,1), B=(7,1), C=(7,7), D=(1,7). For AC/BD, the two segment regions share (4,4). Any line putting A,C strictly in one open half-plane puts their whole segment there; the same argument puts BD in the other. The common point would have to lie strictly on both sides, a contradiction. No separator exists.

For AB/CD, horizontal y=4 separates them strictly; any horizontal line with 1<y<7 works. Many other tilted lines also work. No dot may be on the line and no unique separator is claimed.

Adult general certificate: if finite nonempty hulls K,L are disjoint, compactness gives closest points P in K,Q in L at positive distance. Set d=Q-P. A small move from P toward any X in K cannot reduce the squared distance to Q, so (X-P) dot d<=0; similarly (Y-Q) dot d>=0 for Y in L. Hence K lies on the P side of the perpendicular bisector of PQ and L on the Q side, with strict positive gap. Conversely convex open half-planes make any strict separator rule out shared hull points. This proof is context; the printed task uses the crossing-point explanation and a concrete line.

Hold-back hints

Ask whether a successful line would separate the whole regions, not only the four printed dots. If attempts keep missing, draw both group segments and look for a point that every separator would have to separate from itself.

Depth and further questions

Touching hulls still cannot be strictly separated: their touching point gives the same contradiction. In a disjoint example, moving a successful line slightly while the positive gap remains preserves separation, explaining why many separators can exist. Do not replace strict separation by a line carrying some dots without stating a new rule.

Facilitator notes

The colors/groups are fixed in this task, unlike the base search for a successful partition. Keep the two groupings on separate tracing copies of one unchanged square.

Problem 4-6: One common point for three groups

Entry, prerequisites and timing

P4 is a concrete line entry for grades 2-3; P5 adds a spatial construction. P6's complete impossibility argument is a grades 4-5 return. Count labels and retain three groups, not three pairwise intersection marks. Allow 20-35 minutes or more than one visit.

Mathematical facts and solutions

P4 top row positions are 1,2,4,5,7. AE/BD/C and AD/BE/C both share C at position 4. They are the two successful unordered splits: with five distinct positions, any successful three-group split must have pair,pair,singleton sizes; two singleton groups would disagree. The singleton must be the middle point so two remaining pairs can each straddle it. For bottom row P,Q,R,S, every split has at least two distinct singleton groups; their point regions cannot share one point.

The sharp line theorem: any five ordered positions x1<=x2<=x3<=x4<=x5 use outer pair {x1,x5}, inner pair {x2,x4}, and singleton {x3}. All contain x3, including repeated locations. Four distinct positions give the counterexample, so five is the smallest guaranteed number on a line.

P5 the regular hexagon is centered at (4,4), radius 3.5 units with equal axis scaling. AD,BE,CF join opposite vertices and share the center. All six labels are used once. The independent partition audit finds this is the only success, but finding one is all the task requests.

P6 uses the strictly convex irregular hexagon with P1's coordinates. Every singleton is an extreme vertex outside the hull of the others, so no successful split can have a singleton; the three groups must be pairs. If three disjoint-endpoint chords share a point, each crosses the other two. Around a convex hexagon, each chord must leave one endpoint of each other pair on either side, forcing opposite pairs AD,BE,CF. But AD and BE meet at (37/9,28/9), while CF at that x has y=179/72, not 28/9. Thus even this only candidate fails, and no split works.

AD/BE/CF do meet pairwise, at three different points; that is insufficient. A checkable alternative is the finite audit of all fifteen pairings, after eliminating singleton-containing partitions. No general planar seven-point theorem is asserted.

Hold-back hints

For P4, ask where a singleton's point would have to lie inside both other groups. For P5, try long segments across the hexagon. For P6, first test whether any extreme dot can be a singleton; then retain the possible paired chords and check one common point instead of accepting three pairwise crossings.

Depth and further questions

On a line, r>=2 nonempty groups are guaranteed to share a point with 2r-1 labels: use a middle singleton and r-1 nested outer pairs. With 2r-2 distinct positions, any r-group split has at least two singleton groups, since at most one singleton would require at least 1+2(r-1)=2r-1 labels. Their distinct points cannot meet. This proves the sharp line threshold; no higher-dimensional formula is inferred.

Facilitator notes

The exact convexity and crossing checks are mathematical evidence, not a physical-overlay rehearsal. A child may first conjecture the irregular board impossible and return later for the complete argument. Preserve the regular success beside it for contrast.

Sources, checks and classroom record

Sources and adaptation

Current Week 22 base outline and guide pp. 3-4 and 20 were read for closed-hull conventions and exclusions. The base is a four-label, two-group Radon investigation; this encore adds witnesses, fixed-group separation and three-way concurrency.

Gallier and Quaintance, Aspects of Convex Geometry, section 3.5, is the base guide's recorded Radon context. Its source body was not newly read here. The triangulation, compact-hull separation and sharp line proofs above were independently derived.

Math Circle by the Bay, Preface pp. vii-x (local PDF pp. 8-11), actually read: interaction, manipulatives and long development of deep themes. Our target-removal launch and selected-band returns are proposed adaptations.

Verification

Independent student checks used exact rational hull/orientation arithmetic for all singletons, pairs and triples at T, all 25 five-line and six four-line partitions, and all 90 three-group partitions of each hexagon. Every diagram's equal scaling and the regular hexagon were checked. The author mapped the actual final six tasks and rederived guide claims.

The final guide received a separate independent mathematical review. Both final PDFs were rendered and inspected and passed a clean extracted-source ZIP rebuild. See release-validation.json in the editable source package. No tracing registration, marker use, physical polygon inspection or classroom pilot has occurred.

What to record after a visit

Record problems used, kept target witnesses, a separator or obstruction, grouped regions and where a pair wants to return.

Observe whether filled interiors and degenerate regions are retained, whether discarded dots are accidentally required in P1, and whether three pairwise meetings are mistaken for one common point. Keep observed words/actions separate from hypotheses about age fit.

Status

Draft bonus companion. Unpiloted. Digital and mathematical checks do not establish physical fit, procedural rehearsal, staffing success or classroom pacing. Physical pretests have not been performed. Base student pages and base facilitator guide are separate.